

JEE MAINS & IIT JEE ADVANCED

MATHEMATICS

STRAIGHT LINES

DIBY (DO IT BY YOURSELF)

By: Sachin Jakhar (Defender Mamu)



DIBY-01

If the line $2x + y = k$ passes through the point which divides the line segment joining the points $(1, 1)$ and $(2, 4)$ in the ratio 3: 2, then k equals. [AIEEE-2012]

- (A) $\frac{29}{5}$ (B) 5 (C) 6 (D) $\frac{11}{5}$

[Ans.: (C)]

DIBY-02

The number of points having both coordinates as integers, that lie in the interior of the triangle with vertices $(0, 0)$, $(0, 41)$ and $(41, 0)$ is [JEE Mains-2015]

- (A) 901 (B) 861 (C) 820 (D) 780

[Ans.: (D)]

DIBY-03

Let k be an integer such that triangle with vertices $(k, -3k)$, $(5, k)$ and $(-k, 2)$ has area 28 sq. units. Then the orthocentre of this triangle is at the point [JEE Mains-2017]

- (A) $\left(2, \frac{1}{2}\right)$ (B) $\left(2, -\frac{1}{2}\right)$ (C) $\left(1, \frac{3}{4}\right)$ (D) $\left(1, -\frac{3}{4}\right)$

[Ans.: (A)]



DIBY-04

Let the orthocentre and centroid of a triangle be $A(-3, 5)$ and $B(3, 3)$, respectively. If C is the circumcentre of this triangle, then the radius of the circle having line segment AC as diameter, is **[JEE Mains-2018]**

- (A) $\frac{3\sqrt{5}}{2}$ (B) $\sqrt{10}$ (C) $2\sqrt{10}$ (D) $3\sqrt{\frac{5}{2}}$

[Ans.: (D)]

DIBY-05

If the vertices P, Q, R of a $\triangle PQR$ are rational points, which of the following points of the $\triangle PQR$ is/are always rational point(s) **[IIT-JEE 1988]**

(A rational point is a point both of whose coordinates are rational numbers)

- (A) centroid (B) incentre (C) circumcentre (D) orthocentre

[Ans.: (A), (C), (D)]

DIBY-06

The incentre of the triangle with vertices $(1, \sqrt{3})$, $(0, 0)$ and $(2, 0)$ is **[IIT-JEE 2000]**

- (A) $(1, \sqrt{3}/2)$ (B) $(2/3, 1/\sqrt{3})$ (C) $(2/3, \sqrt{3}/2)$ (D) $(1, 1/\sqrt{3})$

[Ans.: (D)]



DIBY-07

Find the locus of a point such that the sum of its distance from the points $(0, 2)$ and $(0, -2)$ is 6.

$$[\text{Ans. : } 9x^2 + 5y^2 = 45]$$

DIBY-08

Find the locus of the point $(t^2 - t + 1, t^2 + t + 1), t \in R$.

$$[\text{Ans. : } x = \left(\frac{y-x}{2}\right)^2 - \left(\frac{y-x}{2}\right) + 1]$$

DIBY-09

If $P = (1, 0)$, $Q = (-1, 0)$ and $R = (2, 0)$ are three given points, then locus of the points satisfying the relation, $SQ^2 + SR^2 = 2SP^2$ is [IIT-JEE-1998]

- | | |
|--|---|
| (A) a straight line parallel to X -axis | (B) a circle passing through the origin |
| (C) a circle with the centre at the origin | (D) a straight line parallel to Y -axis |

$$[\text{Ans. : (D)}]$$

DIBY-10

The number of integer values of m , for which the x -coordinate of the point of intersection of the lines $3x + 4y = 9$ and $y = mx + 1$ is also an integer, is [IIT-JEE 2001]

- (A) 2 (B) 0 (C) 4 (D) 1

[Ans.: (A)]

DIBY-11

The vertices of a triangle are $A(-1, -7)$, $B(5, 1)$ and $C(1, 4)$. The equation of the bisector of the angle ABC is ____ [IIT-JEE 1993]

[Ans.: $7y = x + 2$]

DIBY-12

Lines $L_1: y - x = 0$ and $L_2: 2x + y = 0$ intersect the line $L_3: y + 2 = 0$ at P and Q respectively. The bisector of the acute angle between L_1 and L_2 intersects L_3 and R .

Statement I: The ratio $PR: RQ$ equals $2\sqrt{2}: \sqrt{5}$.

Statement II: In any triangle, the bisector of an angle divides the triangle into two similar triangles. **[IIT-JEE 2007]**

- (A) Statement I is true, Statement II is also true; Statement II is the correct explanation of Statement I
- (B) Statement I is true, Statement II is also true; Statement II is not the correct explanation of Statement I
- (C) Statement I is true; Statement II is false
- (D) Statement I is false; Statement II is true

[Ans. : (c)]

DIBY-13

Let $A(-2, -1)$, $B(1, 0)$, $C(\alpha, \beta)$ and $D(\gamma, \delta)$ be the vertices of a parallelogram $ABCD$. If the point C lies on $2x - y = 5$ and the point D lies on $3x - 2y = 6$, then the value of $|\alpha + \beta + \gamma + \delta|$ is equal to ____ . **[JEE Mains-2024]**

[Ans. : 32]



DIBY-14

If the line segment joining the points $(5, 2)$ and $(2, a)$ subtends an angle $\frac{\pi}{4}$ at the origin, then the absolute value of the product of all possible values of a is: **[JEE Mains-2024]**

- (A) 4 (B) 8 (C) 6 (D) 2

[Ans.: (A)]

DIBY-15

Let PS be the median of the triangle with vertices $P(2, 2)$, $Q(6, -1)$ and $R(7, 3)$. The equation of the line passing through $(1, -1)$ and parallel to PS is

- (A) $2x - 9y - 7 = 0$ (B) $2x - 9y - 11 = 0$
(C) $2x + 9y - 11 = 0$ (D) $2x + 9y + 7 = 0$

[Ans.: (D)]

DIBY-16

Let $C(\alpha, \beta)$ be the circumcenter of the triangle formed by the lines $4x + 3y = 69$, $4y - 3x = 17$, and $x + 7y = 61$. Then $(\alpha - \beta)^2 + \alpha + \beta$ is equal to **[JEE Mains-2023]**

- (A) 15 (B) 17 (C) 16 (D) 18

[Ans.:]

DIBY-17

Let a variable line of slope $m > 0$ passing through the point $(4, -9)$ intersect the coordinate axes at the points A and B . The minimum value of the sum of the distances of A and B from the origin is **[JEE Mains-2024]**

- (A) 30 (B) 15 (C) 10 (D) 25

[Ans.: (D)]

DIBY-18

The equations to the straight lines which join the origin and the points of trisection of the portion of the line $x + 3y - 12 = 0$ intercepted between the axes of co-ordinates, is

- (A) $y = \frac{2}{3}x$ (B) $y = \frac{x}{6}$ (C) $y = \frac{x}{3}$ (D) $y = \frac{4}{3}x$

[Ans.: (A), (B)]

DIBY-19

If the orthocentre of the triangle, whose vertices are $(1, 2)$, $(2, 3)$ and $(3, 1)$ is (α, β) , then the quadratic equation whose roots are $\alpha + 4\beta$ and $4\alpha + \beta$ is **[JEE Mains-2023]**

(A) $x^2 - 19x + 90 = 0$

(B) $x^2 - 18x + 80 = 0$

(C) $x^2 - 22x + 120 = 0$

(D) $x^2 - 20x + 99 = 0$

[Ans.: (D)]

DIBY-20

A straight line cuts off the intercepts $OA = a$ and $OB = b$ on the positive directions of x-axis and y-axis respectively. If the perpendicular from origin O to this line makes an angle $\frac{\pi}{6}$ with positive direction of y-axis and the area of $\triangle OAB$ is $\frac{98}{3}\sqrt{3}$, then $a^2 - b^2$ is equal to **[JEE Mains-2023]**

(A) $\frac{392}{3}$

(B) 196

(C) $\frac{196}{3}$

(D) 98

[Ans.: (A)]

DIBY-21

The portion of the line $4x + 5y = 20$ in the first quadrant is trisected by the lines L_1 and L_2 passing through the origin. The tangent of an angle between the lines L_1 and L_2 is:

[JEE Mains-2024]

- (A) $8/5$ (B) $25/41$ (C) $2/5$ (D) $30/41$

[Ans.: (D)]

DIBY-22

In a triangle PQR , the co-ordinates of the point P and Q are $(-2, 4)$ and $(4, -2)$ respectively. If the equation of the perpendicular bisector of PR is $2x - y + 2 = 0$, then the circumcenter of the $\triangle PQR$ is

[JEE Mains-2021]

- (A) $(-1, 0)$ (B) $(1, 4)$ (C) $(0, 2)$ (D) $(-2, -2)$

[Ans.: (D)]

DIBY-23

If the triangle ABC has vertices $A(-1, 7)$, $B(-7, 1)$ & $C(5, -5)$, then find the orthocentre of this triangle. **[JEE Mains-2020]**

- (A) $(-3, 3)$ (B) $\left(-\frac{3}{5}, \frac{3}{5}\right)$ (C) $(3, -3)$ (D) $\left(\frac{3}{5}, -\frac{3}{5}\right)$

[Ans.: (A)]

DIBY-24

If (α, β) is the orthocentre of the triangle ABC with vertices $A(3, -7)$, $B(-1, 2)$ and $C(4, 5)$, then $9\alpha - 6\beta + 60$ is equal to **[JEE Mains-2023]**

- (A) 25 (B) 35 (C) 30 (D) 40

[Ans.: (A)]

DIBY-25

If the orthocentre of the triangle formed by the lines $2x + 3y - 1 = 0$, $x + 2y - 1 = 0$ and $ax + by - 1 = 0$, is the centroid of another triangle, whose circumcentre and orthocentre respectively are $(3, 4)$ and $(-6, -8)$, then the value of $|a - b|$ is **[JEE Mains-2024]**

[Ans.: 16]



DIBY-26

Two vertices of a triangle are $(5, -1)$ and $(-2, 3)$. If origin is the orthocentre of the triangle, then the third vertex is

- (A) $(4, 7)$ (B) $(-4, 7)$ (C) $(4, -7)$ (D) $(-4, -7)$

[Ans. : (D)]

DIBY-27

Two vertices of a triangle ABC are $A(3, -1)$ and $B(-2, 3)$ and its orthocentre is $P(1, 1)$. If the coordinates of the point C are (α, β) and the centre of the circle circumscribing the triangle PAB is (h, k) , then the value of $(\alpha + \beta) + 2(h + k)$ equals

[Ans. : 5]

DIBY-28

The centroid of an equilateral triangle is $(2, -1)$ and the equation of its base is $x + 2y = 1$. The length of its sides is

- (A) $\frac{1}{\sqrt{5}}$ (B) $\frac{2\sqrt{5}}{\sqrt{3}}$ (C) $\frac{\sqrt{3}}{\sqrt{5}}$ (D) $\frac{2\sqrt{3}}{\sqrt{5}}$

[Ans. : (D)]

DIBY-29

Lines are drawn parallel to the line $4x - 3y + 2 = 0$, at a distance $3/5$ from the origin. Then which one of the following points lies on any of these lines? **[JEE Mains-2019]**

- (A) $\left(-\frac{1}{4}, -\frac{2}{3}\right)$ (B) $\left(-\frac{1}{4}, \frac{2}{3}\right)$ (C) $\left(\frac{1}{4}, -\frac{1}{3}\right)$ (D) $\left(\frac{1}{4}, \frac{1}{3}\right)$

[Ans. : (B)]

DIBY-30

Let (α, β) be the centroid of the triangle formed by the lines $15x - y = 82$, $6x - 5y = -4$ and $9x + 4y = 17$. Then $\alpha + 2\beta$ and $2\alpha - \beta$ are the roots of the equation **[JEE Mains-2023]**

- (A) $x^2 - 7x + 12 = 0$ (B) $x^2 - 14x + 48 = 0$
(C) $x^2 - 13x + 42 = 0$ (D) $x^2 - 10x + 25 = 0$

[Ans. : (C)]

DIBY-31

If point $(a^2, a + 1)$ lies in the angle between the line $3x - y + 1 = 0$ and $x + 2y - 5 = 0$ containing the origin, then find the value of a .

[Ans. : $a \in (-3, 0) \cup \left(\frac{1}{3}, 1\right)$]



DIBY-32

Two consecutive sides of a parallelogram are $4x + 5y = 0$ and $7x + 2y = 0$. If the equation of one diagonal is $11x + 7y = 0$, find the equation of the other diagonal.

[Ans. : $y - x = 0$]

DIBY-33

A ray of light coming from the point $P(1, 2)$ gets reflected from the point Q on the x -axis and then passes through the point $R(4, 3)$. If the point $S(h, k)$ is such that $PQRS$ is a parallelogram, then hk^2 is equal to:

[JEE Mains-2024]

- (A) 70 (B) 80 (C) 60 (D) 90

[Ans. : (A)]

DIBY-34

In a triangle PQR , the co-ordinates of the points P and Q are $(-2, 4)$ and $(4, -2)$ respectively. If the equation of the perpendicular bisector of PR is $2x - y + 2 = 0$, then the circumcenter of the $\triangle PQR$ is

[JEE Mains-2021]

- (A) $(-1, 0)$ (B) $(1, 4)$ (C) $(0, 2)$ (D) $(-2, -3)$

[Ans. : (D)]



DIBY-35

$A(2, -1), B, C$ are the vertices of a triangle. The equations of the bisector of $\angle ABC$, and of the altitude from B , are $3x - 2y + 5 = 0$ and $7x - 10y + 1 = 0$, respectively Find the equation of BC .

[Ans. : $5x + y + 17 = 0$]

DIBY-36

Let a ray of light passing through the point $(3, 10)$ reflects on the line $2x + y = 6$ and the reflected ray passes through the point $(7, 2)$. If the equation of the incident ray is $ax + by + 1 = 0$, then $a^2 + b^2 + 3ab$ is equal to ____

[JEE Main-2024]

[Ans. : 1]

DIBY-37

Let the point $P(\alpha, \beta)$ be at a unit distance from each of the two lines $L_1 : 3x - 4y + 12 = 0$ and $L_2 : 8x + 6y + 11 = 0$. If P lies below L_1 and above L_2 , then $100(\alpha + \beta)$ is equal to

[JEE Mains-2022]

(A) -14

(B) 42

(C) -22

(D) 14

[Ans. : (D)]

DIBY-38

Let $A(a; b)$, $B(3, 4)$ and $C(-6, -8)$ respectively denote the centroid, circumcentre and orthocentre of a triangle. Then, the distance of the point $P(2a + 3, 7b + 5)$ from the line $2x + 3y - 4 = 0$ measured parallel to the line $x - 2y - 1 = 0$ is **[JEE Mains-2024]**

- (A) $\frac{17\sqrt{5}}{6}$ (B) $\frac{15\sqrt{5}}{7}$ (C) $\frac{17\sqrt{5}}{7}$ (D) $\frac{\sqrt{5}}{17}$

[Ans. : (C)]

DIBY-39

A ray of light coming from the point $P(1, 2)$ gets reflected from the point Q on the x -axis and then passes through the point $R(4, 3)$. If the point $S(h, k)$ is such that $PQRS$ is a parallelogram, then hk^2 is equal to: **[JEE Mains-2024]**

- (A) 60 (B) 70 (C) 80 (D) 90

[Ans. : (B)]

DIBY-40

A light rays emits from the origin making an angle 30° with the positive x-axis. After getting reflected by the line $x + y = 1$, if this ray intersects x-axis at Q , then the abscissa of Q is **[JEE Mains-2023]**

- (A) $\frac{2}{(\sqrt{3}-1)}$ (B) $\frac{2}{3-\sqrt{3}}$ (C) $\frac{\sqrt{3}}{2(\sqrt{3}+1)}$ (D) $\frac{2}{3+\sqrt{3}}$

[Ans. : (D)]

DIBY-41

A ray of light along $x + \sqrt{3}y = \sqrt{3}$ gets reflected upon reaching x-axis, the equation of the reflected ray is **[JEE Mains-2013]**

- (A) $y = x + \sqrt{3}$ (B) $\sqrt{3}y = x - \sqrt{3}$
 (C) $y = \sqrt{3}x - \sqrt{3}$ (D) $\sqrt{3}y = x - 1$

[Ans. : (B)]

DIBY-42

The point $P(a, b)$ undergoes the following three transformations successively

- (a) reflection about the line $y = x$
- (b) translation through 2 units along the positive direction of x -axis.
- (c) rotation through angle $\frac{\pi}{4}$ about the origin in the anti-clockwise direction.

If the co-ordinates of the final position of the point P are $\left(-\frac{1}{\sqrt{2}}, \frac{7}{\sqrt{2}}\right)$ then the value of $2a + b$ is equal to: [JEE Mains-2021]

- (A) 5 (B) 7 (C) 13 (D) 9

[Ans. : (D)]

DIBY-43

In a triangle PQR , the co-ordinates of the points P and Q are $(-2, 4)$ and $(4, -2)$ respectively. If the equation of the perpendicular bisector of PR is $2x - y + 2 = 0$, then the circumcenter of the $\triangle PQR$ is

[JEE Mains-2021]

- (A) $(-1, 0)$ (B) $(1, 4)$ (C) $(0, 2)$ (D) $(-2, -2)$

[Ans. : (D)]



DIBY-44

$A(2, -1), B, C$ are the vertices of a triangle. The equations of the bisector of $\angle ABC$, and of the altitude from B , are $3x - 2y + 5 = 0$ and $7x - 10y + 1 = 0$, respectively Find the equation of BC .

[Ans. : $5x + y + 17 = 0$]

DIBY-45

The equations of two sides AB and BC of a triangle ABC are $4x + y = 14$ and $3x - 3y = 5$, respectively. The point $\left(2, -\frac{4}{3}\right)$, divides the third side BC internally in the ratio $2 : 1$, the equation of the side BC is

[JEE Mains-2024]

(A) $x + 3y + 2 = 0$ (B) $x - 6y - 10 = 0$ (C) $x - 3y - 6 = 0$ (D) $x + 6y + 6 = 0$

[Ans. : (A)]

DIBY-46

Find the distance, measure parallel to $4x - 3y = 1$, of the point $(-5, -7)$ from the bisector of obtuse angle between the lines $3x - 2y = 1$ & $4x - 6y + 2 = 0$?

[Ans. : 10]



DIBY-47

A is $(4, 1)$, B is $(0, 4)$, point P on the line $y = 3x - 1$ is such that $|PA - PB|$ is maximum is

- (A) $(5, 2)$ (B) $(2, 5)$ (C) $(3, 5)$ (D) none

[Ans. : (B)]

DIBY-48

Line L is $3x + 2y + 10 = 0$, $A(5, 2)$, $B(3, 4)$, then point P on line L for which $|PA - PB|$ is maximum is

- (A) $(-24, 31)$ (B) $(-4, 11)$ (C) $(11, -4)$ (D) None

[Ans. : (A)]

DIBY-49

If a point $P(h, k)$ lies on the line $3x + 2y + 10 = 0$ such that $|PA - PB|$ is maximum where A is $(5, 2)$ & B is $(3, 4)$. Find the value of $(h + k)$.

[Ans. : 7]



DIBY-50

A ray of light incident along the line $x - 2y - 3 = 0$ and strikes a line mirror. If the equation of normal on the line mirror at the point of incidence is $2x + 3y + 1 = 0$, then find the equation of the reflected ray

[Ans. : $29x - 2y - 31 = 0$]

DIBY-51

Mixed term xy is to be removed from the general equation $ax^2 + by^2 + 2hxy + 2gx + 2fy + c = 0$, one should rotate the axes through an angle θ given by $\tan 2\theta =$

(A) $\frac{a-b}{2h}$

(B) $\frac{2h}{a+b}$

(C) $\frac{a+b}{2h}$

(D) $\frac{2h}{a-b}$

[Ans. : (D)]

DIBY-52

If the slope of one of the lines given by $a^2x^2 + 2hxy + b^2y^2 = 0$ be three times of the other, then $|h|$ is equal to

- (A) $2\sqrt{3} ab$ (B) $-2\sqrt{3} ab$ (C) $\frac{2}{\sqrt{3}} ab$ (D) $-\frac{2}{\sqrt{3}} ab$

[Ans. : (c)]

DIBY-53

If the line $y = x\sqrt{3}$ cuts the curve $x^3 + y^3 + 3xy + 5x^2 + 3y^2 + 4x + 5y - 1 = 0$ at the points A, B and C , then $OA \cdot OB \cdot OC$ (where O is origin) is

- (A) $\frac{12\sqrt{3}-4}{13}$ (B) $\frac{3\sqrt{3}+12}{13}$ (C) $\frac{4(3\sqrt{3}-1)}{13}$ (D) $\frac{4\sqrt{3}-3}{13}$

[Ans. : (c)]



DIBY-54

If $2x + 3y = 7$ makes equal angles with pair of lines $9x^2 + 12xy + ky^2 = 0$ then k equals

- (A) 7 (B) 14 (C) -14 (D) -7

[Ans. : (B)]

DIBY-55

The area of the triangle formed by the angle bisectors of the pair of lines $x^2 - y^2 + 2y - 1 = 0$ and the line $x + y = 3$ (in sq. units) is

[IIT-JEE-2004]

- (A) 1 (B) 2 (C) 3 (D) 4

[Ans. : (B)]

DIBY-56

Let PQR be right angled isosceles triangle, right angled at $P(2, 1)$. If the equation of the line QR is $2x + y = 3$, then the equation representing the pair of lines PQ and PR is

[IIT-1999]

(A) $3x^2 - 3y^2 + 8xy + 20x + 10y + 25 = 0$ (B) $3x^2 - 3y^2 - 8xy - 20x - 10y + 25 = 0$

(C) $3x^2 - 3y^2 + 8xy + 10x + 15y + 20 = 0$ (D) $3x^2 - 3y^2 + 8xy - 10x - 15y - 20 = 0$

[Ans.: (B)]

DIBY-57

The point of intersection of the lines represented by the equation

$$2x^2 + 3y^2 + 7xy + 8x + 14y + 8 = 0$$

(A) $(0, 2)$ (B) $(1, 2)$ (C) $(-2, 0)$ (D) $(-2, 1)$

[Ans.:]

DIBY-58

The line $lx + my + n = 0$ intersects the curve $ax^2 + 2hxy + by^2 = 1$ at the point P and Q . The circle on PQ as diameter passes through the origin. Prove that $n^2(a + b) = l^2 + m^2$.

DIBY-59

If the ratio of gradients of the lines represented by $ax^2 + 2hxy + by^2 = 0$ is $1:3$, then the value of the ratio $h^2:ab$ is

- (A) $1/3$ (B) $3/4$ (C) $4/3$ (D) 1

[Ans.: (C)]

DIBY-60

The equation of pair of lines joining origin to the points of intersection of $x^2 + y^2 = 27$ and $x + y = 3$ is

- (A) $x^2 + y^2 + 6xy = 0$ (B) $x^2 + y^2 + 3xy = 0$
(C) $xy = 0$ (D) None

[Ans.: (B)]

Column-I

- (A) The orthocentre of the triangle formed by the lines $xy = 0$ and $x + y = 1$ is
- (B) The incentre of the triangle with vertices $(1, \sqrt{3})$, $(0, 0)$ and $(2, 0)$ is
- (C) Let $O(0, 0)$, $P(3, 4)$ and $Q(6, 0)$ be the vertices of the $\triangle OPQ$. The point R inside the $\triangle OPQ$ is such that the triangles OPR , PQR , OQR are of equal area. Then the coordinates of R are
- (D) If $P(1, 2)$, $Q(4, 6)$, $R(5, 7)$ and $S(a, b)$ are the vertices of a parallelogram $PQRS$, then (a, b) equals

Column-II

- (p) $\left(1, \frac{1}{\sqrt{3}}\right)$
- (q) $(0, 0)$
- (r) $(2, 3)$
- (s) $\left(3, \frac{4}{3}\right)$

Ans: A $\rightarrow$ (q), B $\rightarrow$ (p), C $\rightarrow$ (s), D $\rightarrow$ (r)

DIBY-62

Let $A(-1, 1)$, $B(3, 4)$ and $C(2, 0)$ be given three points. A line $y = mx, m > 0$, intersects line AC and BC at a point P and Q respectively. Let A_1 and A_2 be the areas of $\triangle ABC$ and $\triangle PQC$ respectively, such that $A_1 = 3A_2$, then the value of m is equal to

[JEE Mains-2021]

- (A) $\frac{4}{15}$ (B) 1 (C) 2 (D) 3

[Ans: B]

DIBY-63

A triangle ABC lying in the first quadrant has two vertices as $A(1, 2)$ and $B(3, 1)$. If $\angle BAC = 90^\circ$, and $\text{ar}(\triangle ABC) = 5\sqrt{5}$ sq. units, then the abscissa of the vertex C is:

[JEE Mains-2020]

- (A) $1 + \sqrt{5}$ (B) $1 + 2\sqrt{5}$ (C) $2\sqrt{5} - 1$ (D) $2 + \sqrt{5}$

[Ans: B]

DIBY-64

Consider the triangles with vertices $A(2, 1)$, $B(0, 0)$ and $C(t, 4)$, $t \in [0, 4]$. If the maximum and the minimum perimeters of such triangles are obtained at $t = \alpha$ and $t = \beta$ respectively, then $6\alpha + 21\beta$ is equal to

[JEE Mains-2023]

[Ans: 48]

DIBY-65

Let the area of the triangle with vertices $A(1, \alpha)$, $B(\alpha, 0)$ and $C(0, \alpha)$ be 4 sq. units. If the points $(\alpha, -\alpha)$, $(-\alpha, \alpha)$ and (α^2, β) are collinear, then β is equal to

- (A) 64 (B) -8 (C) -64 (D) 512

[JEE Mains-2022]

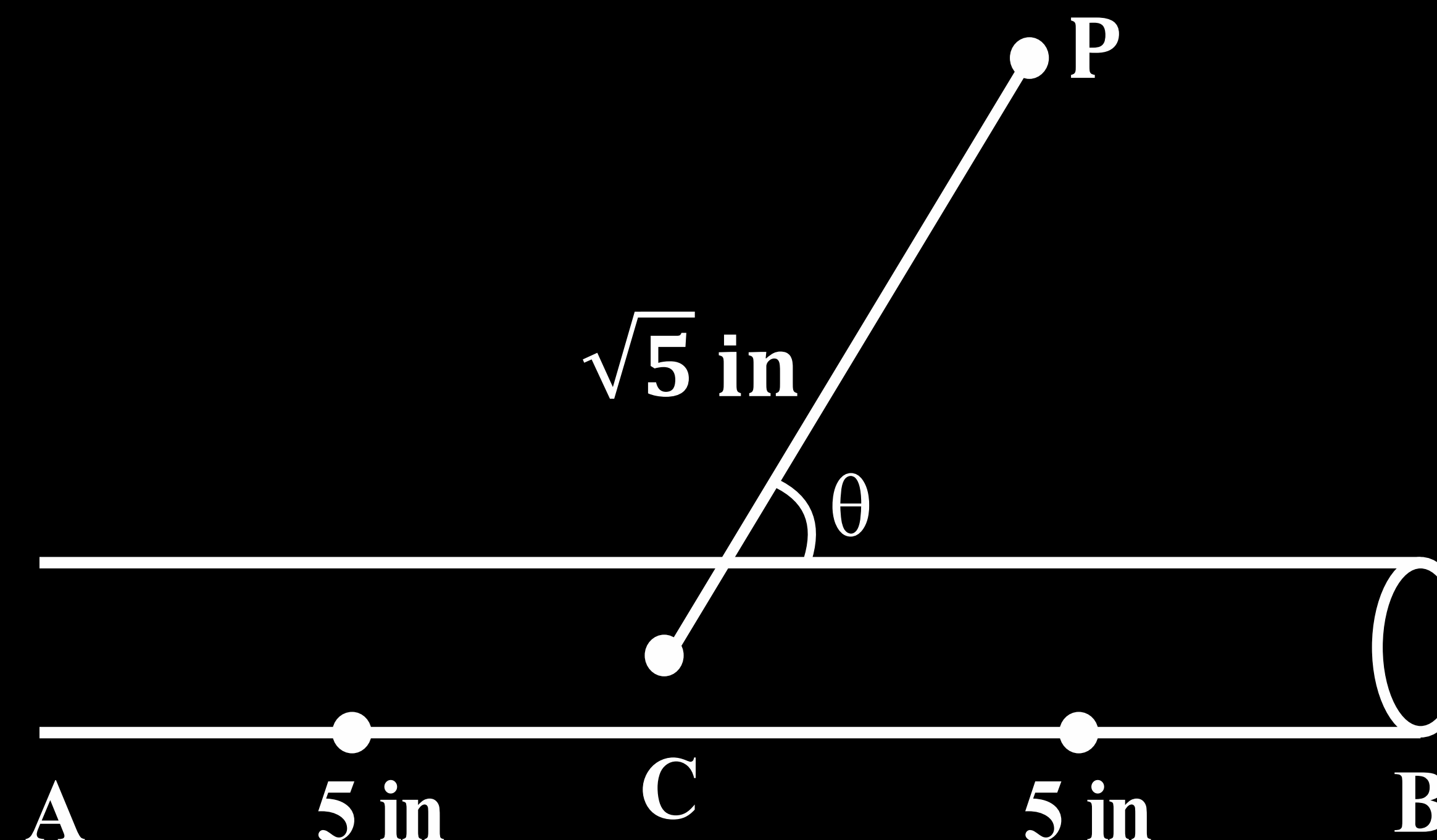
[Ans: C]

DIBY-66

A 10 inches long pencil AB with mid point C and a small eraser P are placed on the horizontal top of a table such that $PC = \sqrt{5}$ inches and $\angle PCB = \tan^{-1}(2)$. The acute angle through which the pencil must be rotated about C so that the perpendicular distance between eraser and pencil becomes exactly 1 inch is:

[JEE Mains-2021]

- (A) $\tan^{-1}\left(\frac{1}{2}\right)$
 (B) $\tan^{-1}\left(\frac{4}{3}\right)$
 (C) $\tan^{-1}(1)$
 (D) $\tan^{-1}\left(\frac{3}{4}\right)$



[Ans: D]

DIBY-67

A point on the straight line, $3x + 5y = 15$ which is equidistant from the coordinate axes will lie only in:

[JEE Mains-2019]

- (A) 1st and 2nd quadrants
(C) 1st, 2nd and 4th quadrant

- (B) 4th quadrant
(D) 1st quadrant

[Ans: A]

DIBY-68

A straight line L at a distance of 4 units from the origin makes positive intercepts on the coordinate axes and the perpendicular from the origin to this line makes an angle of 60° with the line $x + y = 0$. Then an equation of the line L is:

[JEE Mains-2019]

- (A) $(\sqrt{3} + 1)x + (\sqrt{3} - 1)y = 8\sqrt{2}$
(C) $\sqrt{3}x + y = 8$

- (B) $(\sqrt{3} - 1)x + (\sqrt{3} + 1)y = 8\sqrt{2}$
(D) $x + \sqrt{3}y = 8$

[Ans: A]

DIBY-69

Consider a triangle ABC having the vertices $A(1, 2)$, $B(\alpha, \beta)$ and $C(\gamma, \delta)$ and angles $\angle ABC = \frac{\pi}{6}$ and $\angle BAC = \frac{2\pi}{3}$. If the points B and C lie on the line $y = x + 4$, then $\alpha^2 + \gamma^2$ is equal to

[JEE Mains-2024]

[Ans: 14]



DIBY-70

A point P moves so that the sum of squares of its distances from the points $(1, 2)$ and $(-2, 1)$ is 14. Let $f(x, y) = 0$ be the locus of P , which intersects the x -axis at the points A, B and the y -axis at the points C, D . Then the area of the quadrilateral $ACBD$ is equal to
[JEE Mains-2022]

(A) $\frac{9}{2}$

(B) $\frac{3\sqrt{17}}{2}$

(C) $\frac{3\sqrt{17}}{4}$

(D) 9

[Ans: B]

DIBY-71

The equations of the sides AB, BC and CA of a triangle ABC are: $2x + y = 0, x + py = 21a, (a \neq 0)$ and $x - y = 3$ respectively. Let $P(2, a)$ be the centroid of $\triangle ABC$. Then $(BC)^2$ is equal to
[JEE Mains-2023]

[Ans: 122]

DIBY-72

A man is walking on a straight line. The arithmetic mean of the reciprocals of the intercepts of this line on the coordinate axis is $\frac{1}{4}$. Three stones, A, B and C are placed at the point $(1, 1), (2, 2)$ and $(4, 4)$ respectively. Then which of these stones is/are on the path of the man.
[JEE Mains-2021]

(A) B only

(B) A only

(C) All of three

(D) C only

[Ans: A]

DIBY-73

Let $O(0,0)$ and $A(0,1)$ be two fixed points. Then the locus of a point P such that the perimeter of $\triangle AOP$ is 4, is **[JEE Mains-2019]**

(A) $8x^2 - 9y^2 + 9y = 18$

(B) $9x^2 + 8y^2 - 8y = 16$

(C) $8x^2 + 9y^2 - 9y = 18$

(D) $9x^2 - 8y^2 + 8y = 16$

[Ans. : (B)]

DIBY-74

If the locus of the point, whose distances from the point $(2, 1)$ and $(1, 3)$ are in the ratio $5 : 4$, is $ax^2 + by^2 + cxy + dx + ey + 170 = 0$, then the value of $a^2 + 2b + 3c + 4d + e$ is equal to **[JEE Mains-2024]**

(A) 5

(B) -27

(C) 37

(D) 437

[Ans. : (C)]



*It is never about **end result**
It's always about the **Journey***

THANKYOU

#future **IITians**